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Impact and importance of electrostatic potential calculations for predicting structural patterns of hydrogen and halogen bonding†

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The way in which differences in electrostatic potential values can govern how ditopic hydrogen/halogen-bond acceptors form supramolecular interactions in the solid state has been examined using two heterocycles, [1,2,3]triazalo-[3,5-α]quinoline and, [1,2,3]triazalo-[3,5-α]pyridine. In general, the "better" donor (as ranked by electrostatic potential values) preferentially binds to the acceptor atom that carries the higher electrostatic potential value but the outcome is crucially dependent on the magnitude of the potential-energy differences, and structural selectivity can be fine-tuned to generate specifically targeted supramolecular architectures. DFT calculations on [1,2,3]triazalo-[3,5-α]quinoline and [1,2,3]triazalo-[3,5-α]pyridine indicate that the differences in electrostatic potential between the competing sites on the molecules are 38 kJ mol^{-1} and 26 kJ mol^{-1}, respectively. In order to establish how the nature of the electrostatic potential differences impact the structural landscape around these molecules both heterocycles were co-crystallized with twelve hydrogen-bond (HB) donors, five halogen-bond (XB) donors, and four mixed HB/XB donors. A total of 42 new co-crystals were obtained (as determined by IR spectroscopy), eight of which yielded crystals suitable for single-crystal X-ray diffraction. Results indicate that an electrostatic potential difference greater than 38 kJ mol^{-1} leads to a pronounced intermolecular preference for the best acceptor but no selectivity is obtained with a smaller difference. This study shows that electrostatic factors play important roles for defining the structural landscape around molecules capable of engaging in a numerous and competing intermolecular interactions as long as the difference between potentially competing sites is large enough.

Introduction

Hydrogen bonds are frequently employed in applied synthetic supramolecular chemistry.[1] The deliberate use of such interactions as a synthetic tool has produced numerous discrete or extended supramolecular architectures through desired molecular recognition events,[2–5] thus providing a foundation for practical crystal engineering.[6–9] A key aspect of advanced crystal engineering centres on our ability to simultaneously employ several different (and potentially competing) non-covalent interactions in a predictable manner which essentially means that we need to minimize "synthon crossover".[10] Over two decades ago, Etter proposed that "the best proton donors and acceptors remaining after intermolecular hydrogen-bond formation form intermolecular hydrogen bonds to one another"[11] but this 'rule-of-thumb' requires that we have access to an effective and general method for ranking different donors sites with respect to each other. A generally valuable approach for ranking acceptor/donor sites involves using calculated molecular electrostatic potential (MEP) values[12–15] and this method relies on the assumption that the relatively long-range effect of electrostatic interactions play a key role in the initial stages of molecular recognition and assembly events that eventually lead to stable supramolecular entities.[16–19] In several recent studies calculated MEPs have been employed successfully for predicting the most likely outcome among competing hydrogen bonds in a variety of co-crystals.[12,19]

Halogen bonds, which have many fundamental similarities to hydrogen bonds[20–24] have also found practical applications in areas where hydrogen bonds have been successfully employed including catalysis,[25–28] crystal engineering,[29] drug design,[30] solid state synthesis[31] and many others.[32] Thus, it is not surprising that the electrostatic component[33] of halogen bonding makes these interactions amenable to similar MEP based rankings and binding preferences.[16,34]

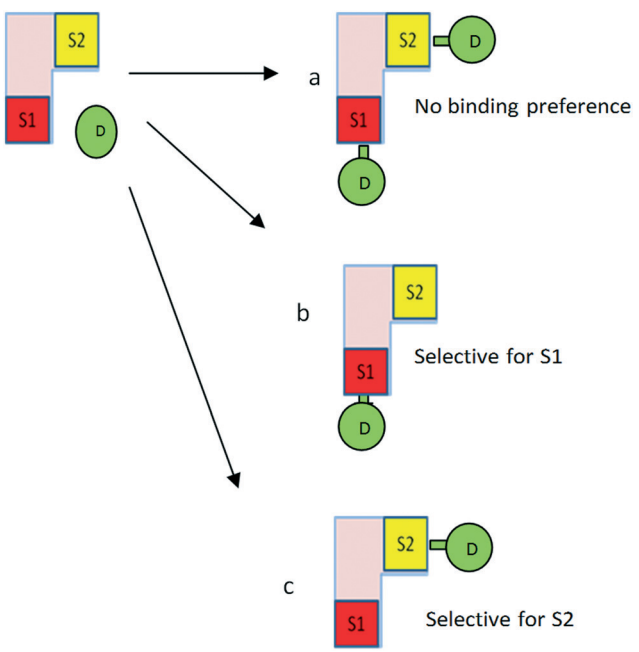
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† Electronic supplementary information (ESI) available: FT-IR grinding experiment data table, crystallographic data for the crystals and crystallography experiment details. CCDC 1505712–1505720. For ESI and crystallographic data in CIF or other electronic format see DOI: 10.1039/c6ce02089e

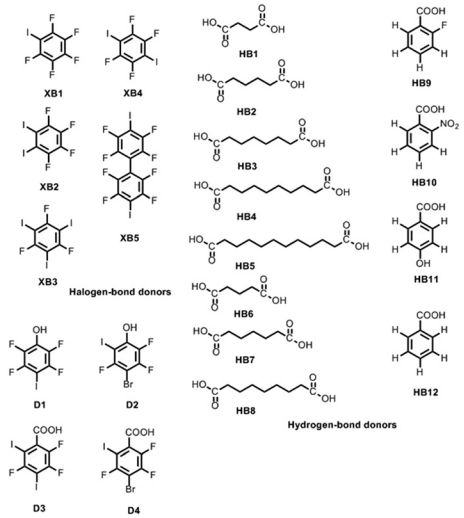
Even though calculated MEPs can be used for predicting the outcome of competing non-covalent interactions we still do not have enough data to allow us to determine more accurately where the exact boundaries are between a precise and selective interaction and assemblies dictated by other weaker and often isotropic contributions. Previous reports have indicated that a difference greater than 75 kJ mol^{-1} in electrostatic potential between two competing sites will lead to selectivity, whereas a difference less than 30 kJ mol^{-1} is too small to produce reliable directed-assembly. In this study, we aim to explore the structural chemistry of molecules located closer to these borders with a view to improving the 'resolution' of our understanding of practical crystal engineering. The choice of target compounds, [1,2,3]triazalo-[3,5-α]quinoline and [1,2,3]triazalo-[3,5-α]pyridine, was determined by the relative difference of the electrostatic potential of their potentially competing binding sites, and in Scheme 1 we illustrate how three different structural outcomes can be envisioned when a dissymmetric ditopic acceptor is introduced to a hydrogen- or halogen-bond donor (Scheme 2).

Experimental

Quinoline-2-carbaldehyde and pyridine-2-carbaldehyde were purchased from Aldrich and utilized without further purification. Hydrogen bond donors **HB1–HB12**, halogen bond donors **XB1–XB5** and mixed donors **D1** and **D2** were purchased from commercial sources and used as received. Mixed donor **D3** and **D4** (Scheme 2) were prepared according to reported procedures.[35] [1,2,3]triazalo-[3,5-α]quinoline and [1,2,3]-triazalo-[3,5-α]pyridine were synthesized following literature



Scheme 1 Postulated structural permutations when multitopic acceptors (**S1**, **S2**) interact with a suitable donor.



Scheme 2 Halogen-bond donors, hydrogen-bond donors, and mixed donors.

data.[36,37] Melting points were measured using Fisher-Johns melting point apparatus. ^{1}H NMR was reported using Varian Unity plus 400 MHz spectrophotometer in $CDCl_3$. IR Spectroscopy was carried out on a Nicolet 380 FT-IR.

Synthesis of [1,2,3]triazalo-[3,5-α] quinolone (A1)[36]

A mixture of quinolone-2-carboxaldehyde (17.5 mmol, 1.375 g) and hydrazine hydrate (206 mmol, 5.2 g, 5 ml) was heated to reflux in methanol (50 ml). Upon completion of the reaction (4 h), the mixture was quenched with an aqueous solution of NaOH (20 ml, 30%) and the resulting mixture was extracted with dichloromethane (3 × 50 ml). The organic extracts were combined, washed with brine (20 ml) and dried over anhydrous $MgSO_4$. Mixture was filtered and concentrated to get the corresponding hydrazone. The hydrazone was directly diluted in chloroform (20 ml) then activated manganese dioxide (42 mmo 1.85 g) was added, heated to reflux (12 h) the resulting mixtrure was cooled to room temperature and filtered with celite. After concentrating [1,2,3]triazalo-[3,5-α] quinoline was obtained as a yellow solid. 65% yield. Mp: 135–140 °C (Lit value 144 °C).[36]

Synthesis of [1,2,3]triazalo-[3,5-α]pyridine (A2)[37]

A mixture of pyridine-2-carboxaldehyde (17.5 mmol, 1.375 g) and hydrazine hydrate (206 mmol, 5.2 g, 5 ml) was heated to reflux in methanol (50 ml). Upon completion of the reaction (4 h), the mixture was quenched with an aqueous solution of NaOH (20 ml, 30%) and the resulting mixture was extracted with dichloromethane (3 × 50 ml). The organic extracts were combined, washed with brine (20 ml) dried over anhydrous $MgSO_4$. The mixture was filtered and concentrated to get the corresponding hydrazone. The hydrazone was diluted in chloroform (20 ml) then activated manganese dioxide (42 mmol 1.85 g) was added, heated to reflux (12 h) the resulting mixture was cooled to room temperature and filtered with

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celite. After concentrating [1,2,3]triazalo-[3,5-α]pyridine was obtained as a yellow solid. 65% yield. Mp: 32–34 °C. (Lit value: 34–39 °C).[37]

Electrostatic potential calculations

Molecular electrostatic potential surfaces for **A1** and **A2** were obtained using density functional theory (B3LYP) using 6-31G* basis set in vacuum. All calculations were carried out using the Spartan'08 software, Fig. 1.

Synthesis of co-crystals

A1 and **A2** were initially screened for co-crystal formation[38] using solvent assisted grinding[39] against twelve hydrogen bond donors, five halogen bond donors and four mixed donors (Scheme 2). All acceptors and donors were ground together with a drop of methanol according to the stoichiometry and the resulting solids were analyzed *via* IR spectrometry. Two broad bands around 1850–2450 cm^{-1}, indicative of intermolecular O–H···N hydrogen bonds,[40] (Fig. 2) were used as primary markers for hydrogen-bond driven co-crystals and whereas co-crystals with halogen bonds were confirmed by significant changes of several IR stretches in the ground mixture compared to the spectra of the pure acceptor and donor compounds.

All positive combinations were then subjected to crystal growth by dissolving 10 mg of the ground mixture in minimum amount of solvent (usually in 1–2 ml of methanol) followed by slow evaporation at ambient conditions. If this method did not produce X-ray quality crystals other methods like vapour diffusion and solvent diffusion were tried, Table 1, and altogether eight new crystal structures were obtained.

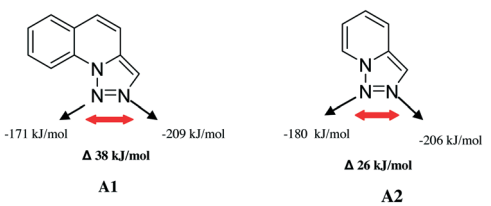


Fig. 1 MEPS difference of two acceptor sites on **A1** and **A2**.

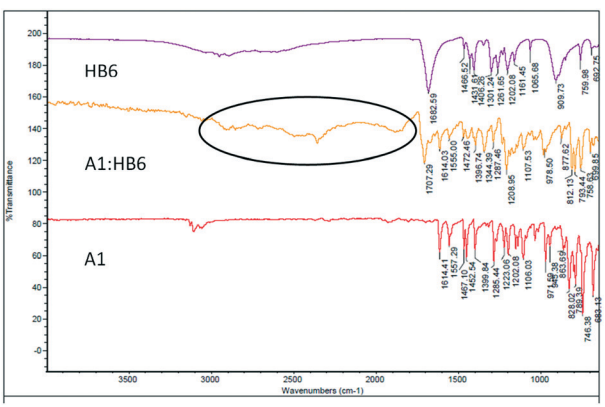


Fig. 2 IR spectrum of **A1:HB6** (middle) with broad bands indicating intermolecular O–H···N interactions (black circle).

Table 1 Experimental details of obtained co-crystals

Crystal	Melting point (°C)	Method used	Solvents used
A1:XB1	85–90	Slow evaporation	Ethyl acetate
A1:XB4	78–84	Slow evaporation	Ethyl acetate
A1:HB2	100–110	Vapour diffusion	Methylene chloride/hexane
A1:HB6	75–80	Slow evaporation	Methanol
A1:HB9	62–70	Slow evaporation	Methanol
A1:D3	115–120	Slow evaporation	Methylene chloride
A2:XB2	55–60	Vapour diffusion	Methylene chloride/hexane
A2:XB3	105–110	Slow evaporation	Methanol

Results and discussion

Acceptor molecules **A1** and **A2** both contain two possible acceptor sites. Molecular electrostatic potential surface calculations (MEPs) showed that the two acceptor sites of **A1** has −171 kJ mol^{-1} and −209 kJ mol^{-1} electrostatic potential giving rise to 38 kJ mol^{-1} potential difference. Similarly, **A2** has values of −180 kJ mol^{-1} and −206 kJ mol^{-1} respectively leading to a 26 kJ mol^{-1} difference. (Fig. 1) The results from the co-crystal screening using infrared spectroscopy as the analytical tool indicate that 33 of the 42 attempted reactions resulted in a co-crystal. Even though IR can provide unambiguous confirmation of co-crystal formation, it does not allow us to determine how the two reactants interact which is where crystallography comes into play; six co-crystals with **A1** and two co-crystals with **A2**, Table 1, and yielded crystals suitable for single-crystal X-ray diffraction.

The crystal structure determination of **A1:XB1** (Fig. 3) shows that the two reactants are present in a 1 : 1 stoichiometry. The halogen bond that drives the co-crystal formation involves the activated iodine atom (as the donor) and the nitrogen atom with the highest electrostatic potential as the acceptor atom (I···N 3.0662(17) Å and C–I···N 167.49 (6)°).

With a ditopic halogen-bond donor **A1** produced a co-crystal **A1:XB4** in the expected 2 : 1 stoichiometry. The two symmetry-related acceptors reside on either side of **XB4** and, as in the previous structure, the best acceptor (as ranked by electrostatic potential values) acts as the halogen-bond acceptor (I···N 2.9379(14) Å and C–I···N 175.98(5)°, (Fig. 4).

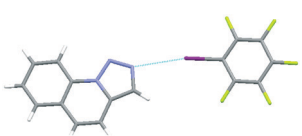


Fig. 3 Primary halogen-bond interaction in the crystal structure of **A1:XB1**.

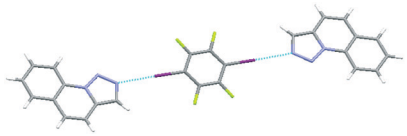


Fig. 4 Primary halogen bonds in the crystal structure of **A1:XB4**.

In the crystal structures of **A1:HB2** and **A1:HB6**, which both involved an aliphatic dicarboxylic acid, the outcome was a co-crystal with the expected 2 : 1 stoichiometry. Again, in both cases the nitrogen atom involved in the key intermolecular interaction was the best acceptor (O⋯N 2.7022(13) Å OH⋯N 172.5(14)° and O⋯N 2.679(3) Å OH⋯N 175(3)° respectively), Fig. 5.

In the crystal structure of **A1:HB9**, involving an aromatic monotopic carboxylic acid, (Fig. 6) the result is the expected 1 : 1 stoichiometry and the O–H hydrogen-bond donor is again interacting with the best acceptor site of **A1** (O⋯N 2.6544(15) Å, OH⋯N 173.2(16)°. The last crystal structure of **A1** was obtained with a mixed XB/HB donor, **D1**, and the outcome is a co-crystal in a 2 : 1 stoichiometry (Fig. 7). Both the hydrogen bond and the halogen bond involve the best acceptor site on **A1**. (O⋯N 2.725(3) Å, 2.751(3) Å OH⋯N 175.1°, 173.1° I⋯N 2.8739(22) Å, 2.8657(22) Å CI⋯N 174.55(37), 173.65(76)°).

The best acceptor atom on **A1** (based on calculated MEPs) was involved in the primary intermolecular interaction in all six crystal structures obtained which underscores the importance of MEPs for predicting outcomes of potentially competing non-covalent interactions. However, it is possible that this supramolecular selectivity was aided by, or a direct result of, some geometric factor which prevented the second nitrogen atom (the second-best acceptor) from participating. With this in mind, we decided to examine the crystal structure of **A1** by itself, (crystals were obtained through slow evaporation from ethyl acetate). The intermolecular interactions found in the structure of **A1** include an C–H⋯N (best acceptor) and a C–H⋯N (second-best acceptor) hydrogen bond which suggests that there is no inherent geometric barrier in this molecule that would prevent a hydrogen (or halogen) bond from taking place with either nitrogen atom (Fig. 8). Consequently, the fact that the same nitrogen atom is engaged in a key structure-directing interaction in all six co-crystals of **A1** can be rationalized against a background of calculated MEP.

To investigate the possible impact of the donor : acceptor ratio of reactants in solution on the chemical composition of the resulting co-crystal, we carried out co-crystallizations from solutions with an acceptor : donor ratio of 1 : 2 and 1 : 3 for **A1:XB2**, **A1:HB2** and **A1:HB9**. The resulting solids were harvested and analyzed *via* ^{1}HNMR which consistently showed the same 2 : 1 acceptor : donor ratio that had previously been found using single-crystal X-ray diffraction. Co-crystals of 1,2,3-[3,5-α]-triazalopyridine, **A2**, a ditopic acceptor with a MEP difference of less than 30 kJ mol^{-1} between the two sites were then examined in order to explore if such a difference is sufficient to impart supramolecular selectivity. Two co-crystals of **A2** gave crystals suitable for single-crystal X-ray diffraction, (Fig. 9).

In the structures of **A2:XB2** and **A2:XB3** both nitrogen atoms participated in halogen bonds with the ditopic donors which indicate that the difference in electrostatic potential in this case is not large enough to produce a molecule with supramolecular structural selectivity.

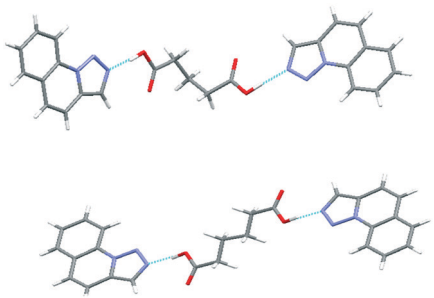


Fig. 5 Main hydrogen bonds in the crystal structures of **A1:HB6** (top) and **A1:HB2** (bottom).

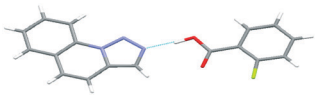


Fig. 6 Hydrogen bond in the crystal structure of **A1:HB9**.

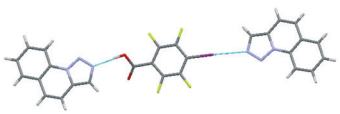


Fig. 7 Primary hydrogen- and halogen bonds in the crystal structure of **A1:D1**.

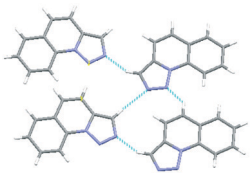


Fig. 8 Hydrogen bonds in the crystal structure of **A1**.

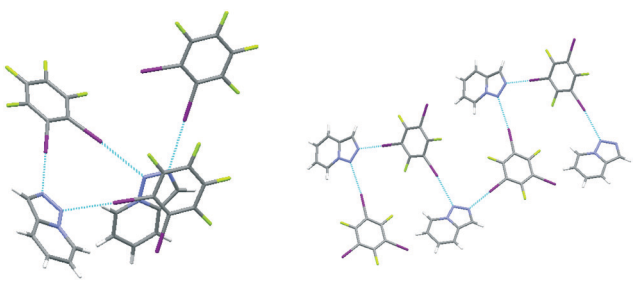
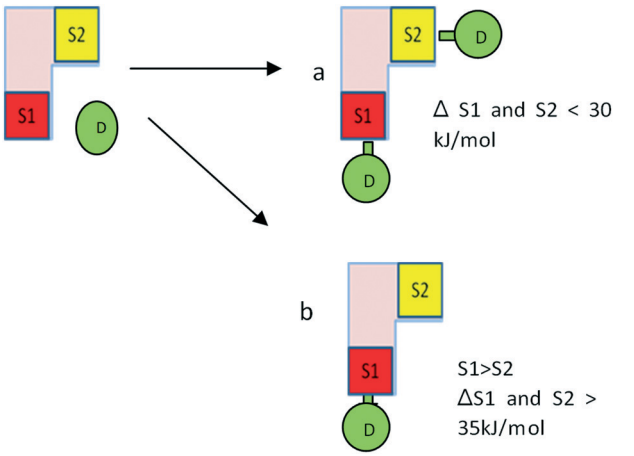


Fig. 9 Primary halogen-bond interactions in the crystal structures of **A2:XB2** (I⋯N 2.988(3)Å, 2.907(4) Å CI⋯N 171.93(13)° 179.53(13)°) and **A2:XB3**, (I⋯N 2.845(3) Å, 2.992(3) Å CI⋯N 177.31(10)° 173.32(11)°) respectively.

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Scheme 3 Overview of structural outcomes.

Conclusions

Two ditopic hydrogen/halogen-bond acceptor molecules were co-crystallized with a series of co-formers in order to identify the magnitude of MEP difference that is needed to achieve a selective structural assembly process. For **A1**, with a ΔMEP value between the two nitrogen atoms of 38 kJ mol^{-1}, six co-crystals were obtained and in each case, the best acceptor atom was exclusively participating in a single hydrogen/halogen bond. In contrast, when the ΔMEP value dropped to 26 kJ mol^{-1}, as was the case for **A2**, the structural selectivity was lost and both acceptor sites were simultaneously engaged in halogen bonds. A crystal structure determination of **A1** (which showed that both nitrogen atoms can be available for short intermolecular contacts at the same time) suggests that geometric or steric explanations for the observed supramolecular preferences are highly unlikely. Instead, a simple electrostatic approach for ranking different interaction sites offers reliable guidelines for predicting the outcome of competing non-covalent interactions, (Scheme 3). It should be pointed out that an extensive polymorph screen was not carried out for each supramolecular reaction and it is certainly possible that new polymorphs with alternative synthon arrangements can be found in the structural landscape surrounding **A1** and **A2**. However, given the fact that the best acceptor (as ranked by its MEP value) was targeted by HB-donors, XB-donors as well as by mixed XB/HB donor molecules, it seems unlikely that the molecular recognition preferences displayed by **A1**/**A2** will change drastically even if several new structures were to be added. In future studies we will include the possible role of solvent and we will also explore a broader range of di- and tritopic acceptors as well as donors to develop robust guidelines for reliable co-crystal synthesis.

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